YOUSSEF KABBARY

AI Engineer | Computer Vision | Deep Learning

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github.com/YoussefKabbary | linkedin.com/in/youssef-kabbary | youssefkabbary.github.io | Military Service: Exempted

PROFESSIONAL SUMMARY

AI graduate (B.Sc. Computer Science, Artificial Intelligence, Pharos University, 2025) who ships working, measured AI systems: real-time computer vision (YOLOv8 detection, pose, tracking, ReID), a Transformer accident-risk model trained on 191k real crash records, and LoRA fine-tuning of a diffusion model on a 4 GB GPU, each served through FastAPI or Gradio. Cut per-frame inference in a crowd analytics system from 95-232 ms to 43-72 ms. Also designed and delivered MGICO's production ERP end to end (FastAPI, PostgreSQL, 424 automated tests). Currently R&D Engineer at ST Smart Technology. Seeking a Junior AI / Computer Vision Engineer role.

TECHNICAL SKILLS

- **Computer Vision:** Object Detection, Pose Estimation, Multi-Object Tracking (ByteTrack), Image Classification, Semantic Segmentation, Optical Flow, OpenCV
- **Deep Learning:** PyTorch, TensorFlow, YOLOv8 (Ultralytics), Transformers, Diffusion Models, LoRA / PEFT, CNNs (ResNet, MobileNet), Hugging Face, scikit-learn
- **Optimization & Deployment:** FP16 / Mixed Precision, CUDA, ONNX Runtime, Gradient Checkpointing, FastAPI, REST APIs, Gradio, Edge AI (Raspberry Pi)
- **Data & Backend:** Dataset Curation, Model Evaluation (AUC, Baselines), PostgreSQL, SQLAlchemy, pytest
- **Programming:** Python, C++, JavaScript, SQL; Multithreading, Git, GitHub Actions, Linux
- **Cloud:** AWS (studying for the Generative AI Developer - Professional exam)
- **Embedded & IoT:** Arduino, ESP32, Raspberry Pi 5, STM32, Bluetooth HC-05/06, I2C, SPI, PCB (EasyEDA, Proteus)

PROFESSIONAL EXPERIENCE

R&D Engineer | ST Smart Technology, Alexandria | Jun 2026 - Present

- Designed a technical training curriculum from scratch for university students and graduate trainees.
- Deliver hands-on sessions to computer science and engineering students in cohorts of up to 25.
- Run professional training for postgraduate and industry audiences, including PhD holders.
- Mentor student teams building applied engineering projects for technical competitions.
- Built the Smart Aquarium IoT System (v2.0): Arduino Nano controller with temperature, ammonia sensing, automated feeding and Bluetooth monitoring, shipped as a Windows desktop app.
- Built Bolt Control: one Windows app that drives our robots over Bluetooth, with 5 Arduino sketches and CI.

Robotics & AI Instructor (Part-Time) | Genio Tech Robotics, Alexandria | May 2026 - Sep 2026

- Taught an Arduino and robotics curriculum, including a 4-session PCB design course (schematic to PCB, simulation in Proteus and EasyEDA).
- Mentored 20+ students on sensors, line-following robots and autonomous navigation.

Core Team Member, Robotics & AI | Google Developer Student Club (GDSC), Pharos University | Sep 2023 - Jun 2024

- Delivered 8 technical workshops to 100+ students on machine learning, computer vision and robotics (Python, TensorFlow, OpenCV, Arduino).
- Mentored 20+ students on projects involving YOLO object detection and embedded systems.

PROJECTS

ERP System for MGICO Contracting | Client Project, delivered and in daily use | Jul 2026 - Sep 2026

Python, FastAPI, PostgreSQL 17, SQLAlchemy, Pydantic, pytest, JavaScript, Arabic RTL UI, Tailscale

- Commissioned by MGICO, a piling contractor; designed, built and delivered end to end, now running its daily operations: 141 API routes, 43 tables, 28 migrations, 424 automated tests.

- Closed-by-default routes, per-role and per-site access, full audit log, byte-level upload validation.
- Fixed HTTPS link (Tailscale Funnel), self-healing watchdog (36 s outage to recovery), restore-verified backups.

rtl-admin-starter: Arabic RTL Admin Template (FastAPI, JS, pytest) | GitHub | 2026

- Written from scratch, separate from MGICO: RTL-first layout, tables as phone cards, auth gate tested on every route.

Crowd Master: Real-Time Crowd Analytics (v2, v3) | GitHub | 2026

Python, PyTorch, YOLOv8, ByteTrack, ONNX Runtime, OpenCV, FastAPI, CUDA

- Real-time people counting for CCTV, RTSP and webcam feeds: YOLOv8 body detection fused with YOLOv8-pose head positions so partly occluded people still count.
- Doorway entry/exit counting, named-zone occupancy, optical-flow anomaly detection with automatic clip saving, and short-term occupancy forecasting.
- Audited and fixed 28 documented defects: per-frame inference cut from 95-232 ms to 43-72 ms (RTX 3050 Laptop, FP16); gate counting raised from 0/3 to 3/3 correct on an automated regression test.
- v3: watchlist matching and cross-gate person journeys (face, ReID, clothing), gated by measured pixel resolution per person so the system reports what the camera can support instead of silent misses; control-room alerts.

SafeRoute AI: Road-Safety Map & Voice Navigation | saferoute.tail88a5c1.ts.net | 2026

Python, PyTorch, Transformers, YOLOv8, FastAPI, SQLite, OpenStreetMap, Sentinel-2, Leaflet

- Live map of Egypt: fastest vs safest route for trips up to 400 km, every curve measured on an OpenStreetMap graph built on demand, and offline turn-by-turn voice navigation in Arabic.
- Trained a Transformer accident predictor on a case/control dataset built from 191k real NHTSA FARS fatal crashes plus historical weather; location-grouped split tested on 47 unseen areas, AUC 0.628, beating logistic-regression and gradient-boosting baselines.
- Measured 6,026 km of Egyptian roads (785 stretches posted above what they hold); YOLOv8 reads community road photos; 61 live security checks, WCAG 2.2 accessibility checks, 658 tests.

Virtual Try-On: Diffusion Model Fine-Tuning | Independent Project | 2026

Python, PyTorch, CatVTON (Diffusion), LoRA / PEFT, MediaPipe, Gradio

- Fine-tuned a CatVTON diffusion inpainting model with LoRA on a 4 GB laptop GPU (FP16, gradient checkpointing), 3,300 training steps.
- Curated 20,425 images from 25,180 with quality filtering, face tagging and perceptual-hash de-duplication for leakage-free train/validation/test splits.
- Built a Gradio app with pixel-exact face preservation (inner-face MAE 0.0), hand and background locks; about 25 s per image on GPU.

AgliFly: Autonomous Surveillance Drone | Graduation Project, Pharos University | 2025 (Grade: Excellent)

Python, CNN, OpenCV, Raspberry Pi

- CNN-based real-time object detection running on Raspberry Pi for on-board edge inference, integrated with GPS navigation and autonomous flight control.

EDUCATION

B.Sc. Computer Science, Artificial Intelligence | Pharos University in Alexandria | Graduated Sep 2025
Graduation Project: AgliFly autonomous surveillance drone (Grade: Excellent)

CERTIFICATIONS & TRAINING

- AWS Certified Generative AI Developer - Professional (studying for the exam)
- Build with AI: Masr Edition - Google for Developers & ITI, 2026
- Microsoft Learn - Classification models, Facial recognition, Python error handling, 2024
- Robotics (50 h) - European Educational Group, 2024 | Robotics With AI (50 h) - ST Smart, 2023

Tap the link: https://drive.google.com/drive/folders/1GgYbhpk2dvvwPaC4Z8Gs6cVloYViaf8U?usp=drive_link

LANGUAGES

Arabic (Native) | English (B2)